bar. The movement of the eccentric rod away from the cylinder then produces a downward action on the upper arm of the bell crank, causing the valve to come forward. This is just the opposite to the effect of this eccentric rod movement with the reverse yoke in forward motion. The inside admission gear is therefore indirect in back motion; that is, the valve moves in a direction opposite to the movement of the eccentric rod.

MID-GEAR POSITION. The Baker Valve Gear is in mid-position when the centers of the reverse yoke and radius bar bearing bell crank and gear connecting rod bearing, gear connecting rod and radius bar bearing are on a common center line through the middle of the reverse yoke and gear frame pin.

LAP AND LEAD MOVEMENT. The lap and lead movement is obtained from the combination lever simply by proportioning this lever so that when its long arm receives a movement equal to the piston stroke, its short arm will deliver to the valve a movement equal to two times the sum of the lap plus the lead. This movement is separate from and entirely independent of the rest of the valve gear action and therefore remains constant, regardless of the reverse yoke or the reverse lever position.

TO CHANGE THE LEAD. To change the lead of the Baker Valve Gear, it is necessary to change either the lap of the valve or the distance between the connecting points of the combination lever. Reducing the lap of the valve increases the lead, and vice versa. Changing the lap also changes the point of cutoff. The effect of reducing the lap is therefore not only to increase the lead, but also to increase the point of cutoff. The effect of increasing the lap is just the opposite. To increase the lead by means of a change in the combination lever, it is necessary to make the upper arm longer or the lower arm shorter. To decrease the lead these changes are reversed.

RULES. The reverse yoke moves forward in the forward motion.

The eccentric crank follows the main crank pin in the forward motion and leads it in backward motion.

Combination levers for inside admission valves have the valve stem attached below the valve rod connection.

Combination levers for outside admission have the valve stem attached above the valve rod connection.

The Walschaert Valve Gear

Because of its superior advantages and reliability this gear has been widely used on steam locomotives.

Like the Baker Valve Gear, it is an outside gear, taking its motion from an eccentric crank attached to the main crankpin and from the main crosshead.

It consists principally of an eccentric crank, eccentric rod, reverse link, link block radius rod, combination lever and union link, together with the usual reversing mechanism of reach rod, reverse shaft, reverse lever, etc., Fig. 7.

The motion derived from the eccentric crank gives the travel to the valve minus the lap and lead. This motion is transmitted, through a slotted link oscillating on a fulcrum, to the radius rod where it is combined with the motion of the combination lever (lap and lead movement) and thence to the valve by means of the valve stem. Fig.8

The Walschaert Valve Gear

When setting the valves with the Walschaert gear the same initial procedure is followed as outlined under "General Practice".

To simplify the following explanation the examples used to illustrate required changes in the Baker gear will also be used in calculating changes in the Walschaert gear.

Suppose the locomotive on which the valves are to be set has the following specifications:

Inside Admission Valves

Maximum valve travel, 5-3/4"

Eccentric crank throw, 15½"

Constant lead 1/4"

Steam lap of valve 1"

Link block below center in forward gear

The influence of eccentric rod changes on the direction of the movement of the valve is clearly shown in Fig. 11. Here it may be seen that if the eccentric rod E is lengthened to E', the radius rod R will be moved ahead to position R' and the valve stem will be moved the distance X in the direction of the arrow, thus displacing the valve from position V to position V'. From this it may be seen that as in the Baker gear if the eccentric rod is lengthened the valve, in forward motion, will be moved ahead. If the eccentric rod is shortened the valve will be moved back. Obviously the movement of the valve will be reversed in the backward motion, with the link block above center in this case.

Regardless of whether the gear is in forward or backward motion the position of valve travel may be moved ahead by lengthening the radius rod and back by shortening the radius rod.

Bearing these facts in mind let us consider Example No. 1. Following the explanation we find that in this instance no change is necessary in the radius rod of the Walschaert gear. (The radius rod in the Walschaert gear may be considered the same as the valve rod in the Baker gear.)

Referring to Example No. 2, we proceed to determine the amount of error in the valve movement just as explained. However, after the amount of error is found, the correction required in the eccentric rod is somewhat different.

In the Walschaert gear a change in the length of the eccentric rod results in a change in the position of the valve, approximately in proportion to the eccentric throw and valve travel.

In the engine under consideration the eccentric crank throw is 15½" and the valve travel is 5-3/4". The ratio of eccentric throw to valve travel is therefore approximately 2.7 to 1. Hence, the eccentric rod must be shortened 2.7 times 4/64ths, or approximately 10/64 to produce a movement of 1/16 in. in the valve.